The Operator Independence Theorem: π_{sort} is not a function of $R(u)$

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Abstract

We prove that the 0-Hecke sorting operator π_{sort} and the five-vertex Demazure R-matrix $R(u)$ are algebraically independent: no specialization, limit, or conjugation relates them. The proof rests on a minimal polynomial argument—π_{sort} satisfies a degree-2 relation while $R(u)$ has minimal polynomial of degree 4 for all $u \neq 0$. As a corollary, the three-level integrability hierarchy (solvable $\supset$ quasi-solvable $\supset$ combinatorial) consists of genuinely distinct algebraic levels, not degenerations of one another.

1 Definitions

Definition 1 (Ice-rule R-matrix)**.** *We work over $\mathbb{C}$. Let $V = \mathbb{C}^2$ with basis $\{|0\rangle, |1\rangle\}$, and identify $V \otimes V \cong \mathbb{C}^4$ with ordered basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. A* six-vertex *or* ice-rule *R-matrix is a linear map $R\colon V \otimes V \to V \otimes V$ preserving the total spin $i + j$ of each basis vector $|ij\rangle$. In the standard basis it has the form*

$$R = \begin{pmatrix} a_1 & 0 & 0 & 0 \\ 0 & b_1 & c_1 & 0 \\ 0 & c_2 & b_2 & 0 \\ 0 & 0 & 0 & a_2 \end{pmatrix}.$$

Definition 2 (Demazure five-vertex R-matrix)**.** *Set $b_2 = 0$ (five-vertex condition) and*

$$a_1(u) = 1 + u, \quad a_2 = 1, \quad b_1(u) = u, \quad b_2 = 0, \quad c_1 = 1, \quad c_2 = 1,$$

giving the one-parameter family

$$R(u) = \begin{pmatrix} 1+u & 0 & 0 & 0 \\ 0 & u & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1}$$

Definition 3 (0-Hecke sorting operator)**.** *The* sorting operator *$\pi_{\text{sort}}\colon V \otimes V \to V \otimes V$ is defined by*

$$\pi_{\text{sort}}|ij\rangle = \begin{cases} |01\rangle & \text{if } (i,j) = (1,0), \\ |ij\rangle & \text{otherwise}, \end{cases}$$

with matrix form

$$\pi_{\text{sort}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{2}$$

Definition 4 (Braid relation)**.** *A matrix $R \in \mathrm{End}(V \otimes V)$ satisfies the* constant braid relation *if*

$$R_{12}\, R_{23}\, R_{12} = R_{23}\, R_{12}\, R_{23} \tag{3}$$

in $\mathrm{End}(V^{\otimes 3})$*, where* $R_{12} = R \otimes I$ *and* $R_{23} = I \otimes R$*.*

2 The Operator Independence Theorem

Theorem 5 (Operator Independence)**.** *Let $R(u)$ and π_{sort} be as in* (1) *and* (2)*. Then:*

1. π_{sort} ***is a 0-Hecke generator:*** $\pi_{\mathrm{sort}}^2 = \pi_{\mathrm{sort}}$ *and* π_{sort} *satisfies* (3)*.*
2. $R(u)$ ***is not a Hecke generator:*** *For all $u \neq 0$, $R(u)$ has at least three distinct eigenvalues and satisfies no quadratic relation $(R - \alpha I)(R - \beta I) = 0$.*
3. ***No specialization:*** *There is no $u_0 \in \mathbb{C}$ and $c \in \mathbb{C}^*$ such that $R(u_0) = c \cdot \pi_{\mathrm{sort}}$.*
4. ***No limit:*** *There is no function $f \colon \mathbb{C} \to \mathbb{C}^*$ and no $u_0 \in \mathbb{C} \cup \{\infty\}$ such that $\lim_{u \to u_0} R(u)/f(u) = \pi_{\mathrm{sort}}$.*
5. ***No conjugation:*** *There is no $M \in \mathrm{GL}_4(\mathbb{C})$, $u_0 \in \mathbb{C}$, and $c \in \mathbb{C}^*$ such that $M\, R(u_0)\, M^{-1} = c \cdot \pi_{\mathrm{sort}}$.*

Corollary 6 (Integrability Hierarchy)**.** *The three-level hierarchy*

$$\textit{Solvable} \supset \textit{Quasi-solvable} \supset \textit{Combinatorial}$$

consists of genuinely distinct algebraic objects. The quasi-solvable level (π_{sort} satisfying 0-Hecke relations in $H_n(0)$) is not an intermediate degeneration of the solvable level ($R(u)$ satisfying the parametric Yang–Baxter equation).

3 Proof

3.1 Spectral data of $R(u)$

Lemma 7. *$R(u)$ is block-diagonal with respect to the spin decomposition $V \otimes V = V_0 \oplus V_1 \oplus V_2$, where $V_0 = \mathrm{span}\{|00\rangle\}$, $V_1 = \mathrm{span}\{|01\rangle, |10\rangle\}$, $V_2 = \mathrm{span}\{|11\rangle\}$.*

- *On V_0: eigenvalue $\lambda_1 = 1 + u$.*
- *On V_2: eigenvalue $\lambda_2 = 1$.*
- *On V_1: the restricted matrix is $\left(\begin{smallmatrix} u & 1 \\ 1 & 0 \end{smallmatrix}\right)$ with characteristic polynomial $\lambda^2 - u\lambda - 1$, giving eigenvalues*

$$\lambda_\pm = \frac{u \pm \sqrt{u^2 + 4}}{2}. \tag{4}$$

The four eigenvalues of $R(u)$ are therefore $\{1 + u,\ 1,\ \lambda_+,\ \lambda_-\}$.

Proof. The block structure follows from the ice rule (spin conservation). The characteristic polynomial of the V_1 block is $\det\left(\begin{smallmatrix} u-\lambda & 1 \\ 1 & -\lambda \end{smallmatrix}\right) = \lambda^2 - u\lambda - 1$. □

Lemma 8. $\det R(u) = -(1 + u)$.

Proof. $\det R(u) = (1+u)\cdot 1 \cdot \det\left(\begin{smallmatrix} u & 1 \\ 1 & 0 \end{smallmatrix}\right) = (1+u)(0-1) = -(1+u)$. □

Lemma 9. *For $u \neq 0$, the eigenvalue 1 is not a root of $\lambda^2 - u\lambda - 1$.*

Proof. Substituting $\lambda = 1$: $1 - u - 1 = -u \neq 0$. □

Lemma 10. *For $u \neq 0$, the eigenvalue $1+u$ is not a root of $\lambda^2 - u\lambda - 1$.*

Proof. Substituting $\lambda = 1+u$: $(1+u)^2 - u(1+u) - 1 = 1 + 2u + u^2 - u - u^2 - 1 = u \neq 0$. □

Proposition 11. *For $u \neq 0$, the minimal polynomial of $R(u)$ is*

$$m_{R(u)}(\lambda) = (\lambda - (1+u))(\lambda^2 - u\lambda - 1)(\lambda - 1), \tag{5}$$

which has degree 4.

Proof. The minimal polynomial is the least common multiple of the minimal polynomials on each invariant subspace: $(\lambda - (1+u))$ on V_0, $(\lambda^2 - u\lambda - 1)$ on V_1 (the full characteristic polynomial, since the 2×2 block is not scalar for $u \neq 0$), and $(\lambda - 1)$ on V_2. By Lemmas 9 and 10, no pair shares a common root for $u \neq 0$, so their product is their least common multiple, giving degree $1 + 2 + 1 = 4$. □

3.2 Part (1): π_{sort} is a 0-Hecke generator

Idempotency. Direct computation shows $\pi_{\text{sort}}^2 = \pi_{\text{sort}}$. Row 3 of π_{sort} is identically zero, so the only nontrivial entry to check is $(\pi_{\text{sort}}^2)_{23} = (\pi_{\text{sort}})_{22} \cdot (\pi_{\text{sort}})_{23} = 1 \cdot 1 = 1$. All entries match. (Verified symbolically.)

Braid relation. On $V^{\otimes 3}$ with basis $\{|abc\rangle : a, b, c \in \{0,1\}\}$, the operators $(\pi_{\text{sort}})_{12}$ and $(\pi_{\text{sort}})_{23}$ sort adjacent pairs of bits. Both compositions $\pi_{12}\pi_{23}\pi_{12}$ and $\pi_{23}\pi_{12}\pi_{23}$ implement the full *bubble sort* on three bits, sending any triple to its nondecreasing rearrangement. The nontrivial cases are:

Input	$\pi_{12}\pi_{23}\pi_{12}$	$\pi_{23}\pi_{12}\pi_{23}$	Output
$\|100\rangle$	$\to \|010\rangle \to \|001\rangle \to \|001\rangle$	$\to \|100\rangle \to \|010\rangle \to \|001\rangle$	$\|001\rangle$
$\|110\rangle$	$\to \|110\rangle \to \|101\rangle \to \|011\rangle$	$\to \|101\rangle \to \|011\rangle \to \|011\rangle$	$\|011\rangle$
$\|101\rangle$	$\to \|011\rangle \to \|011\rangle \to \|011\rangle$	$\to \|101\rangle \to \|011\rangle \to \|011\rangle$	$\|011\rangle$

The remaining five inputs are already sorted and fixed by both sides. (Verified by 8×8 matrix computation.) □

3.3 Part (2): $R(u)$ is not a Hecke generator for $u \neq 0$

A matrix satisfies a quadratic relation $(R - \alpha I)(R - \beta I) = 0$ if and only if its minimal polynomial has degree at most 2. By Proposition 11, for $u \neq 0$ the minimal polynomial of $R(u)$ has degree 4. Therefore no quadratic relation holds. □

Remark 12. At $u = 0$, the R-matrix is the permutation (swap) P with $P^2 = I$ and eigenvalues $\{1, 1, 1, -1\}$. This satisfies the quadratic $(R - I)(R + I) = 0$, making it a generator of the symmetric group algebra $\mathbb{C}[S_n]$—the *combinatorial* level of the hierarchy.

3.4 Part (3): No specialization

Suppose $R(u_0) = c \cdot \pi_{\text{sort}}$ for some $u_0 \in \mathbb{C}$, $c \in \mathbb{C}^*$. Comparing the $(1,1)$ and $(2,2)$ entries:

$$R(u_0)_{11} = 1 + u_0 = c, \qquad R(u_0)_{22} = u_0 = c.$$

Subtracting: $(1 + u_0) - u_0 = 1 = 0$, a contradiction.

Remark 13. The obstruction is structural: the diagonal entries of $R(u)$ are $1 + u$ and u, which differ by a constant additive shift of 1. Those of π_{sort} are both 1. No scalar rescaling can absorb an additive discrepancy.

□

3.5 Part (4): No limit

We show that for no $u_0 \in \mathbb{C} \cup \{\infty\}$ and no function $f(u)$ does $R(u)/f(u) \to \pi_{\text{sort}}$ as $u \to u_0$.

The key observation: $R(u)$ has entries depending on u (namely $R_{11} = 1 + u$ and $R_{22} = u$) and constant entries ($R_{23} = R_{32} = 1$, $R_{44} = 1$, $R_{33} = 0$). All nonzero entries of π_{sort} equal 1.

Case 1: Finite u_0, $f(u) \to f_0 \neq 0$. Then $R(u)/f(u) \to R(u_0)/f_0$, reducing to Part (3). No solution.

Case 2: Finite u_0, $f(u) \to 0$. The constant entry $R_{23} = 1$ gives $1/f(u) \to \infty$. Diverges.

Case 3: Finite u_0, $f(u) \to \infty$. Then $R_{23}/f(u) \to 0$, but $(\pi_{\text{sort}})_{23} = 1 \neq 0$. Mismatch.

Case 4: $u_0 = \infty$. As $u \to \infty$, the entries $R_{11} = 1 + u$ and $R_{22} = u$ grow without bound while $R_{23} = R_{32} = 1$, $R_{44} = 1$, $R_{33} = 0$ stay bounded. If $f(u) \sim \alpha u^\gamma$ as $u \to \infty$ for $\gamma > 0$:

$$\lim_{u\to\infty} \frac{R(u)}{\alpha u^\gamma} = \begin{cases} \frac{1}{\alpha}\operatorname{diag}(1,1,0,0) & \text{if } \gamma = 1, \\ 0 & \text{if } \gamma > 1, \\ \text{diverges} & \text{if } \gamma < 1. \end{cases}$$

For $\gamma = 1$, the limit has rank 2; π_{sort} has rank 3. No match. □

3.6 Part (5): No conjugation

This is the strongest statement. We use two conjugation invariants: the eigenvalue multiset and the minimal polynomial.

Step 1 (Eigenvalue constraint). If $MR(u_0)M^{-1} = c \cdot \pi_{\text{sort}}$, then $R(u_0)$ and $c \cdot \pi_{\text{sort}}$ share the same eigenvalue multiset. Since π_{sort} has eigenvalues $\{1, 1, 1, 0\}$, the matrix $c \cdot \pi_{\text{sort}}$ has eigenvalues $\{c, c, c, 0\}$: a triple eigenvalue c and a simple eigenvalue 0.

Step 2 (Where is the zero eigenvalue?) From the eigenvalues $\{1 + u, 1, \lambda_+, \lambda_-\}$ of $R(u)$:

- $1 + u = 0$: $u = -1$. (Candidate.)
- $1 = 0$: Impossible.
- $\lambda_\pm = 0$: From (4), $\frac{u \pm \sqrt{u^2+4}}{2} = 0$ requires $u = \mp\sqrt{u^2 + 4}$, hence $u^2 = u^2 + 4$. Impossible.

So $u_0 = -1$ is the unique candidate.

Step 3 (Eliminate $u_0 = -1$). At $u = -1$, the eigenvalues are

$$\left\{0,\ 1,\ \frac{-1 + \sqrt{5}}{2},\ \frac{-1 - \sqrt{5}}{2}\right\}. \tag{6}$$

The three nonzero eigenvalues are 1, $\frac{-1+\sqrt{5}}{2} \approx 0.618$, and $\frac{-1-\sqrt{5}}{2} \approx -1.618$. These are three distinct values. For conjugacy to $c \cdot \pi_{\text{sort}}$, we need all three nonzero eigenvalues to equal c. Since they are distinct, this is impossible.

Step 4 (Alternative: minimal polynomial argument). We give a second proof that works uniformly for all $u \neq 0$. The minimal polynomial of $c \cdot \pi_{\text{sort}}$ is $\lambda(\lambda - c)$ (degree 2), since $(c\pi)^2 = c^2\pi = c(c\pi)$ and $c\pi \neq 0$, $c\pi \neq cI$. By Proposition 11, for all $u \neq 0$ the minimal polynomial of $R(u)$ has degree 4. The minimal polynomial is a conjugation invariant. A matrix with minimal polynomial of degree 4 cannot be conjugate to a scalar multiple of an idempotent (minimal polynomial degree 2).

Step 5 (Eliminate $u_0 = 0$). $R(0) = P$ (the permutation matrix) with eigenvalues $\{1, 1, 1, -1\}$. For conjugacy to $c \cdot \pi_{\text{sort}}$ with eigenvalues $\{c, c, c, 0\}$, we need $-1 = 0$. Impossible.

Combining Steps 2–5: no value of $u_0 \in \mathbb{C}$ admits a conjugacy $MR(u_0)M^{-1} = c \cdot \pi_{\text{sort}}$. □

4 The Conceptual Picture

The proof reveals why the three levels of the integrability hierarchy are algebraically incommensurable. Two independent deformation parameters are at play:

- The **spectral parameter** u, which parametrizes the R-matrix family $R(u)$.
- The **Hecke parameter** q, which controls the quotient from the braid algebra to the Hecke algebra via $(T_i - q)(T_i + 1) = 0$.

Setting different parameters to zero gives:

Limit	Result	Algebra	Relation
$u = 0$ in $R(u)$	Permutation P	$\mathbb{C}[S_n]$	$P^2 = I$ (involution)
$q = 0$ in $H_n(q)$	Sorting π_{sort}	$H_n(0)$	$\pi^2 = \pi$ (idempotent)

These two limits are algebraically incomparable: $P = R(0)$ is an involution with eigenvalues $\{1, 1, 1, -1\}$, while π_{sort} is an idempotent with eigenvalues $\{1, 1, 1, 0\}$. The sign -1 versus 0 is the spectral signature of their algebraic independence.

The "Hecke q" and the "spectral u" control different aspects of the algebraic structure. The sorting operator π_{sort} is the $q = 0$ limit of the Hecke generator—not the $u = 0$ limit of the R-matrix. These are genuinely different operations on genuinely different objects, connected only through the braid group that is a common ancestor of both.

This connects directly to the categorical phase diagram:

- **Braided monoidal** (generic q, generic u): the full Yang–Baxter equation holds; $R(u)$ generates a braid group action.
- **Coboundary monoidal** ($q = 0$): the cactus group replaces the braid group; π_{sort} generates $H_n(0)$ with idempotent relations.
- **Symmetric monoidal** ($u = 0$): the permutation P generates $\mathbb{C}[S_n]$ with involutive relations.

The Operator Independence Theorem says these categorical phases are separated by algebraic walls that no continuous deformation can cross.

5 Computational Verification

All matrix computations were independently verified using SymPy (symbolic) and NumPy (numerical):

- **Part (1):** Idempotency $\pi_{\text{sort}}^2 = \pi_{\text{sort}}$ and braid relation verified by direct 4×4 and 8×8 matrix multiplication (`hecke_integrability_test.py`).
- **Part (2):** Eigenvalues of $R(u)$ computed symbolically; quadratic relation $(R-\alpha I)(R-\beta I) = 0$ tested at $u = 1, 2, \frac{1}{2}, 3$ with no annihilator found (`pi_sort_independence.py`).
- **Part (3):** Entrywise system $R(u_0) = c \cdot \pi_{\text{sort}}$ solved symbolically—empty solution set.
- **Part (4):** Limits computed symbolically for $u \to 0, \pm 1, \infty$ with normalizations $f(u) = 1, u, 1+u, au+b$.
- **Part (5):** Jordan forms and eigenvalue multisets compared at all candidate points $u = 0, -1$.

Scripts: `hecke_integrability_test.py`, `pi_sort_independence.py`.